CERTIFICATE OF RADIOLOGICAL VERIFICATION AND DATASET APPROVAL

This is to certify that the radiological dataset review, cross-sectional landmark validation, pelvimetric data verification, and pelvic floor morphometric assessment for the Ph.D. thesis entitled:

"A COMPUTED TOMOGRAPHY EVALUATION OF PELVIC FLOOR THICKNESS AND BONY DIMENSIONS AMONG WOMEN FROM HILLY AND INLAND REGIONS OF SOUTH INDIA: A COMPARATIVE CROSS-SECTIONAL STUDY"

submitted by Mr. MUTHUKUMAR C, Research Scholar in Anatomy bearing USN No. 24PH14ANA01, Sri Siddhartha Academy of Higher Education (SSAHE), Tumakuru, has been thoroughly reviewed, verified, and counter-checked in the Department of Radiodiagnosis.

The candidate has compiled and processed a structured CT pelvimetry dataset comprising 350 female patient studies - 175 Hilly-region cohort and 175 Inland/Plain-region cohort - sourced from institutional PACS repositories across participating medical institutions in Kerala, Karnataka, and Tamil Nadu. All scans were originally acquired for non-gynaecological clinical indications and were retrospectively screened to satisfy the strict anatomical and demographic inclusion criteria of the study. Each dataset entry includes anonymised DICOM image volumes, mask-segmented bone and soft-tissue structures, and a corresponding CT Pelvimetry & Pelvic Floor Assessment Report (PDF).

I hereby confirm and certify that:

1. **Bony Pelvimetry Validation:** The following linear and angular morphometric parameters were measured at the correct predefined cross-sectional anatomical planes and have been verified for anatomical accuracy: True Conjugate Diameter, Obstetric Conjugate Diameter, Diagonal Conjugate Diameter, Transverse Diameter of the Pelvic Inlet, Interspinous Diameter (ISD), Intertuberous Diameter (ITD), AP Diameter of Mid-pelvis, Subpubic Angle (SPA), Sacral Length, Sacral Curvature, Pelvic Inclination, and Oblique Diameter. All values fall within clinically established reference ranges for the South Indian female population.
2. **Pelvic Floor Soft-Tissue Assessment:** Thickness measurements and functional parameters for the levator ani muscle group (puborectalis, pubococcygeus, and iliococcygeus) and the coccygeus have been reviewed. Bilateral muscle thickness values, levator hiatus dimensions (AP and lateral), hiatal area, resting thickness, Valsalva thickness, and contractility indices have been cross-checked for anatomical

plausibility and regional consistency against published normative data.

3. **Cohort Stratification & PFD Grading:** Each of the 350 study subjects has been correctly categorised by PFD pattern (Descent/Plain-type vs. Bulge/Hilly-type) and graded by severity (Grade 1-3) for the following pelvic floor disorder diagnoses: uterine prolapse, cystocele, rectocele, vault prolapse, and combined PFD. The cohort distribution and clinical grading have been verified as appropriate for comparative cross-sectional analysis.
4. **Segmentation & Landmark Accuracy:** The automated bone and soft-tissue boundary extractions (hip bones, sacrum, pelvic floor musculature) were spot-checked against source DICOM slices across a random sample of cases from both cohorts. Mask bounding-box coordinates, voxel spacing calibrations, and measurement derivations have been counter-verified for technical correctness and reproducibility.
5. **Report Completeness & Clinical Consistency:** Each patient record is accompanied by a structured two-page CT Pelvimetry & Pelvic Floor Assessment Report containing: patient demographics, anthropometric measurements, obstetric history, pelvic floor symptom profile, imaging parameters, full pelvimetry table with normal references, pelvic floor muscle measurements, levator hiatus dimensions, and a radiological impression. All 350 reports have been reviewed and found complete and internally consistent.

The dataset, pelvimetric measurements, soft-tissue morphometric values, and associated clinical reports compiled from the Hilly and Inland study cohorts have been comprehensively audited and are validated as anatomically accurate and radiologically reliable for advanced statistical comparative analysis in fulfilment of the above-mentioned Ph.D. thesis.

Date: ______________________________

Place: ______________________________

Verified and Approved by:

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Signature

Name: ______________________________

MD (Radiodiagnosis)

Designation: ______________________________

Department: ______________________________

Institution: ______________________________

Reg. No.: ______________________________

Official Seal / Stamp